



# Task 02: Security Operations Center (SOC)-Incident Classification , Alert Triage and Incident Response Management

**Cybersecurity Internship – CyArt**

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## 1. Introduction

In modern cybersecurity environments, organizations face thousands of security alerts every day from firewalls, SIEM platforms, intrusion detection systems, antivirus software, and endpoint monitoring solutions. Security Operations Centers (SOCs) are responsible for continuously monitoring these alerts, identifying potential threats, classifying incidents, prioritizing risks, and responding effectively to security events.

The purpose of this task is to develop theoretical and practical knowledge of alert management, incident classification, alert triage, evidence preservation, and incident response processes using industry-standard SOC tools and frameworks. This task focuses on understanding how SOC analysts analyze security incidents, determine their severity levels, validate Indicators of Compromise (IOCs), and document response activities in a structured and professional manner.

During this task, alerts are categorized into severity levels such as Critical, High, Medium, and Low based on factors including business impact, exploit likelihood, asset criticality, and vulnerability severity using the Common Vulnerability Scoring System (CVSS). Incidents are also mapped to frameworks such as MITRE ATT&CK, ENISA Incident Taxonomy, and VERIS to standardize threat classification and improve detection accuracy.

The practical implementation includes configuring and using security tools such as Wazuh, TheHive, CrowdSec, Metasploit, VirusTotal, AlienVault OTX, Velociraptor, and FTK Imager in a virtual SOC lab environment consisting of Kali Linux, Ubuntu Server, and Metasploitable2 virtual machines. Simulated attacks such as brute-force SSH attempts and vulnerable service exploitation are performed to generate real-time alerts for analysis and investigation.

Additionally, this task emphasizes proper incident documentation, escalation procedures, evidence preservation, threat intelligence validation, and post-incident reporting. The complete workflow demonstrates the real-world responsibilities of SOC analysts and provides hands-on experience in cybersecurity monitoring, threat detection, and incident response operations.



## 2. Alert Priority Levels

### Introduction

Alert Priority Levels are used in Security Operations Centers (SOCs) to identify the severity and urgency of security incidents. Since organizations generate thousands of alerts daily from SIEM platforms, intrusion detection systems, firewalls, endpoint protection systems, and network monitoring tools, SOC analysts must prioritize alerts effectively to ensure that critical threats are handled immediately.

Alert prioritization helps SOC analysts:

- reduce alert fatigue
- identify critical threats quickly
- improve incident response efficiency
- minimize business impact
- focus on high-risk incidents

The priority of an alert is determined using factors such as:

- business impact
- exploit likelihood
- asset criticality
- vulnerability severity
- attacker activity
- affected systems

SOC teams generally classify alerts into four categories:

- Critical
- High
- Medium
- Low

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### Core Concepts of Alert Priority Levels

#### 1. Critical Alerts

Critical alerts represent the highest level of severity and indicate active compromise or immediate danger to the organization.



These incidents require immediate response because they can cause:

- complete system compromise
- financial loss
- service disruption
- data breaches
- ransomware encryption

Examples of Critical Alerts

- Ransomware attack
- Domain controller compromise
- Active data exfiltration
- Remote code execution exploit
- Log4Shell exploitation
- Privilege escalation with administrator access

Characteristics

- Active exploitation detected
- High business impact
- Immediate response required
- Potential legal or financial damage

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## 2. High Alerts

High alerts indicate serious suspicious activity that could lead to compromise if not investigated quickly.

Examples of High Alerts

- Malware infection
- Unauthorized administrator login
- Credential dumping
- Suspicious PowerShell execution
- Suspicious outbound traffic

Characteristics

- Requires urgent investigation
- High probability of attack
- Moderate to high business impact



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### 3. Medium Alerts

Medium alerts represent suspicious activities that may indicate attacker reconnaissance or early attack stages.

#### Examples of Medium Alerts

- SSH brute-force attempts
- Multiple failed logins
- Unusual authentication attempts
- Network scanning activity

#### Characteristics

- Requires monitoring and investigation
- May become serious later
- Moderate impact

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### 4. Low Alerts

Low alerts are informational or low-risk activities.

#### Examples of Low Alerts

- Single failed login
- Port scan
- Information gathering attempts
- Minor policy violations

#### Characteristics

- Minimal immediate impact
- Mostly monitoring required
- Useful for threat hunting

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## CVSS Scoring System

CVSS stands for Common Vulnerability Scoring System.



It is used to calculate the severity of vulnerabilities and security incidents.

CVSS scoring helps SOC analysts:

- prioritize vulnerabilities
- assess risk
- improve incident handling
- standardize severity ratings

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CVSS Severity Levels

CVSS Score| Severity

0.0 – 3.9| Low

4.0 – 6.9| Medium

7.0 – 8.9| High

9.0 – 10.0| Critical

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Example – Log4Shell Vulnerability

Vulnerability:

CVE-2021-44228

CVSS Score:

9.8

Severity:

Critical

Reason:

- Remote exploitation possible
- No authentication required
- Full system compromise possible
- Public exploit available

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## Business Impact in Alert Prioritization

SOC teams prioritize alerts based on affected systems.



## Example

System| Priority

Production Server| Critical

Employee Laptop| Medium

Test VM| Low

If ransomware affects a production database server, the alert becomes Critical because it impacts business operations.

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## Exploit Likelihood

SOC analysts also consider exploit availability.

If a vulnerability:

- has public exploits
- is actively exploited
- appears in threat intelligence feeds

then its severity increases.

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## Asset Criticality

Asset criticality refers to the importance of systems in an organization.

Examples:

- Banking server = Critical
- HR database = High
- Test machine = Low

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## MITRE ATT&CK Mapping

Alerts are often mapped to MITRE ATT&CK techniques.

Examples:

Technique ID| Description

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T1566| Phishing  
T1110| Brute Force  
T1190| Exploit Public-Facing Application

This helps SOC analysts understand attacker behavior.

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## Objectives of Alert Prioritization

The main objectives are:

- detect critical threats quickly
- improve SOC efficiency
- reduce response time
- minimize business impact
- improve threat visibility
- standardize alert handling

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## PRACTICAL IMPLEMENTATION – ALERT PRIORITIZATION

Tools Used

Tool| Purpose

Google Sheets| Alert classification

Wazuh| SIEM monitoring

TheHive| Incident ticketing

VirusTotal| IOC validation

AlienVault OTX| Threat intelligence

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### Practical Task 1 – Create Alert Classification Table

Objective

Create a table to classify alerts based on severity and MITRE ATT&CK techniques.

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Step 1 – Open Google Sheets

Create a new spreadsheet.



Rename it:  
Alert Classification System

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## Step 2 – Create Columns

Add the following columns:

Alert ID| Alert Type| Priority| CVSS Score| MITRE Technique| Status

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## Step 3 – Add Mock Alerts

Example data:

| Alert ID | Alert Type        | Priority | CVSS Score | MITRE Technique | Status        |
|----------|-------------------|----------|------------|-----------------|---------------|
| 001      | Log4Shell Exploit | Critical | 9.8        | T1190           | Open          |
| 002      | SSH Brute Force   | Medium   | 5.5        | T1110           | Open          |
| 003      | Phishing Email    | High     | 8.0        | T1566           | Investigating |
| 004      | Port Scan         | Low      | 2.5        | T1046           | Closed        |

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## Practical Task 2 – Alert Prioritization

### Objective

Learn how SOC analysts prioritize alerts using severity and business impact.

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## Step 1 – Analyze Alert Severity

Check:

- CVSS score
- affected system
- attacker activity
- exploit availability

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## Step 2 – Determine Business Impact

Examples:

- Production server compromise = Critical
- Test machine compromise = Medium

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## Step 3 – Assign Priority

Assign:

- Critical
- High
- Medium
- Low

based on investigation findings.

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## Practical Task 3 – Wazuh Dashboard Monitoring

Objective

Visualize security alerts using Wazuh.

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## Step 1 – Open Wazuh Dashboard

Open browser on Ubuntu Server.

Access:

https://YOUR-IP

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## Step 2 – Login

Enter:

- Username
- Password



provided during installation.

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## Step 3 – Navigate to Security Events

Open:

- Security Events
- Threat Hunting
- Dashboard

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## Step 4 – Analyze Alerts

Check:

- authentication failures
- SSH brute-force attempts
- suspicious logins
- malware alerts

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## Step 5 – Create Priority Visualization

Create:

- pie chart
- bar graph

for:

- Critical alerts
- High alerts
- Medium alerts
- Low alerts

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## Practical Task 4 – Incident Ticket Creation in TheHive

Objective



Create and manage incident tickets.

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Step 1 – Open TheHive

Open browser:  
<http://YOUR-IP:9000>

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Step 2 – Login

Enter administrator credentials.

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Step 3 – Create New Case

Click:  
Create Case

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Step 4 – Enter Incident Details

Title:  
[Critical] Ransomware Detected on Server-X

Description:  
Indicators:

- crypto\_locker.exe
- suspicious encryption activity
- malicious IP detected

Priority:  
Critical

Assignee:  
SOC Analyst

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## Step 5 – Save Case

Click:  
Create

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## Practical Task 5 – Escalation Email

### Objective

Escalate critical incidents to Tier 2 analysts.

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### Example Escalation Email

Subject:  
Critical Alert Escalation – Possible Ransomware Activity

### Message:

A critical ransomware-related alert has been detected on Server-X. Multiple suspicious encryption activities and malicious file executions were observed. Initial analysis identified the file crypto\_locker.exe communicating with IP 192.168.1.50. Immediate investigation and containment are recommended to prevent further spread. The incident has been escalated for advanced analysis and response actions.

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### Expected Outcomes

After completing this practical, the SOC analyst should be able to:

- classify alerts correctly
- prioritize incidents effectively
- use CVSS scoring
- map incidents to MITRE ATT&CK
- monitor alerts using Wazuh
- create incident tickets in TheHive
- escalate incidents professionally

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### Conclusion



Alert prioritization is one of the most important responsibilities of a SOC analyst. Proper classification and prioritization allow security teams to respond quickly to critical incidents, reduce business impact, and improve overall security operations. Using tools such as Wazuh, TheHive, VirusTotal, and threat intelligence platforms helps analysts investigate alerts efficiently and improve incident response workflows.

## 3. Incident Classification

### Introduction

Incident classification is one of the most important processes in a Security Operations Center (SOC). It is the process of identifying, categorizing, labeling, and organizing cybersecurity incidents based on their type, severity, behavior, impact, and attack method. Proper incident classification helps SOC analysts investigate threats efficiently, improve incident response speed, and reduce confusion during security investigations.

Organizations receive thousands of alerts daily from firewalls, intrusion detection systems (IDS), antivirus software, endpoint detection tools, SIEM platforms, and cloud security systems. Without proper classification, it becomes difficult for security teams to determine which incidents are dangerous and which are harmless.

Incident classification allows SOC teams to:

- identify attack patterns
- understand attacker behavior
- prioritize threats
- automate investigations
- improve reporting accuracy
- enhance incident response workflows

In modern cybersecurity environments, incident classification is performed using industry-standard frameworks such as:

- MITRE ATT&CK
- ENISA Incident Taxonomy
- VERIS Framework

These frameworks help SOC analysts standardize security event analysis and improve communication between security teams.

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### Core Concepts of Incident Classification



What is a Security Incident?

A security incident is any event that:

- violates security policies
- threatens confidentiality
- affects integrity
- disrupts system availability

Examples include:

- malware infection
- phishing attack
- ransomware attack
- unauthorized access
- brute-force attempts
- data breaches

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## Objectives of Incident Classification

The main objectives are:

- categorize security incidents properly
- improve investigation efficiency
- identify attacker tactics
- reduce incident response time
- improve SOC workflows
- support threat intelligence operations
- improve reporting and documentation

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## Types of Security Incidents

Security incidents are classified into multiple categories based on attack behavior and impact.

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### 1. Malware Incident



Malware refers to malicious software designed to damage, disrupt, or gain unauthorized access to systems.

#### Examples

- Trojan
- Worm
- Spyware
- Ransomware
- Keylogger

#### Indicators

- suspicious processes
- unknown file execution
- abnormal CPU usage
- unauthorized connections

#### Example Scenario

A user downloads a malicious attachment from an email, resulting in ransomware encryption.

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## 2. Phishing Incident

Phishing attacks attempt to steal sensitive information such as:

- passwords
- banking credentials
- OTPs
- confidential data

Attackers commonly use fake emails or malicious websites.

#### Indicators

- suspicious email sender
- fake login pages
- malicious links
- urgent language in email

#### Example Scenario



An employee receives a fake Microsoft login page requesting credentials.

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### 3. DDoS Incident

Distributed Denial of Service (DDoS) attacks flood systems or servers with excessive traffic to make services unavailable.

Indicators

- unusually high network traffic
- server slowdown
- service outage
- repeated requests from multiple IPs

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### 4. Insider Threat Incident

An insider threat occurs when an employee or authorized user misuses access privileges.

Indicators

- unauthorized file access
- abnormal login timings
- unusual data transfers
- privilege misuse

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### 5. Data Exfiltration Incident

Data exfiltration is the unauthorized transfer of sensitive information outside the organization.

Indicators

- large outbound traffic
- cloud uploads
- suspicious external connections
- encrypted file transfers

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## Incident Classification Frameworks





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## MITRE ATT&CK Framework

MITRE ATT&CK is a globally recognized framework used to map attacker tactics and techniques.

It helps SOC analysts:

- identify attacker behavior
- classify attacks
- improve threat detection
- understand attack chains

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## MITRE ATT&CK Tactics

Tactic| Description

Initial Access| How attacker enters system

Execution| Running malicious code

Persistence| Maintaining access

Privilege Escalation| Gaining higher permissions

Credential Access| Stealing credentials

Discovery| Gathering system information

Lateral Movement| Moving across systems

Exfiltration| Stealing data

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## Common MITRE Techniques

Technique ID| Description

T1566| Phishing

T1110| Brute Force

T1190| Exploit Public-Facing Application

T1059| Command and Scripting Interpreter

T1046| Network Service Scanning

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## ENISA Incident Taxonomy

ENISA provides a standardized method for classifying incidents.



It categorizes incidents into:

- malware
- intrusion attempts
- availability attacks
- information gathering
- fraud
- abuse

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## **VERIS Framework**

VERIS stands for:

Vocabulary for Event Recording and Incident Sharing.

It helps organizations:

- document incidents
- share threat intelligence
- standardize reporting

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## **Contextual Metadata in Incident Classification**

SOC analysts enrich incidents using metadata.

Metadata improves investigation quality and helps identify attacker behavior.

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### **Common Metadata Fields**

Field| Description

Source IP| Attacker IP address

Destination IP| Target system

Timestamp| Time of attack

IOC| Indicator of Compromise

Hostname| Affected system

Username| Affected user

File Hash| Suspicious file identifier

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## Indicators of Compromise (IOCs)

IOCs are pieces of evidence showing possible malicious activity.

Examples:

- malicious IP addresses
- suspicious domains
- malware hashes
- phishing URLs
- registry modifications

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## Importance of Incident Classification in SOC

Proper classification helps:

- reduce investigation time
- improve alert prioritization
- improve SOC efficiency
- automate workflows
- improve reporting accuracy
- support compliance requirements

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## PRACTICAL IMPLEMENTATION – INCIDENT CLASSIFICATION

Tools Used

| Tool           | Purpose                 |
|----------------|-------------------------|
| Google Sheets  | Incident classification |
| Wazuh          | Alert monitoring        |
| VirusTotal     | IOC validation          |
| AlienVault OTX | Threat intelligence     |
| TheHive        | Incident management     |

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### Practical Task 1 – Create Incident Classification Table

Objective



Create a table to classify incidents using severity levels and MITRE ATT&CK techniques.

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## Step 1 – Open Google Sheets

Create a new spreadsheet.

Rename:

Incident Classification System

---

## Step 2 – Create Table Columns

Add:

Alert ID| Incident Type| Severity| MITRE Technique| Source IP| Status

---

## Step 3 – Add Sample Incidents

Example:

| Alert ID | Incident Type   | Severity | MITRE Technique | Source IP    | Status        |
|----------|-----------------|----------|-----------------|--------------|---------------|
| 001      | Phishing Email  | High     | T1566           | 192.168.1.10 | Open          |
| 002      | SSH Brute Force | Medium   | T1110           | 192.168.1.20 | Investigating |
| 003      | Port Scan       | Low      | T1046           | 192.168.1.30 | Closed        |

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## Practical Task 2 – Alert Analysis in Wazuh

Objective

Analyze security alerts generated in Wazuh.

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## Step 1 – Open Wazuh Dashboard

Access:

https://YOUR-IP



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## Step 2 – Login

Enter administrator credentials.

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## Step 3 – Open Security Events

Navigate to:

- Security Events
- Threat Hunting

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## Step 4 – Analyze Alert Details

Check:

- source IP
- event time
- alert severity
- affected system
- MITRE mapping

---

## Step 5 – Classify Incident

Determine:

- malware
- brute force
- phishing
- scanning
- suspicious login

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## Practical Task 3 – IOC Validation Using VirusTotal

Objective



Validate suspicious indicators using threat intelligence platforms.

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Step 1 – Open VirusTotal

Search:

- suspicious IP
- file hash
- malicious domain

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Step 2 – Review Detection Results

Check:

- antivirus detections
- malicious score
- community reports

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Step 3 – Document Findings

Record:

- IOC type
- detection ratio
- risk level

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## Practical Task 4 – AlienVault OTX Validation

Objective

Validate attacker IPs and threat intelligence indicators.

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Step 1 – Open AlienVault OTX

Search attacker IP or domain.



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## Step 2 – Review Threat Intelligence

Check:

- reputation
- associated malware
- attack history
- threat pulses

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## Step 3 – Add Findings to Incident Report

Document:

- IOC details
- risk assessment
- associated campaigns

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## Practical Task 5 – Incident Documentation

### Objective

Create professional incident records.

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## Step 1 – Create Incident Report

Include:

- incident summary
- severity
- affected systems
- indicators of compromise
- recommendations

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## Step 2 – Add Investigation Timeline



Example:

Timestamp| Action

10:00 AM| Alert received

10:15 AM| IOC validated

10:30 AM| Incident classified

---

Step 3 – Export Documentation

Save as:

- PDF
- DOCX

Upload to GitHub repository.

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## Expected Outcomes

After completing this practical, the SOC analyst should be able to:

- classify incidents correctly
- map incidents to MITRE ATT&CK
- validate IOCs
- use Wazuh for alert analysis
- improve threat investigations
- create professional documentation

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## Conclusion

Incident classification is an essential SOC process that helps security teams identify, organize, and investigate cyber threats effectively. Using frameworks such as MITRE ATT&CK, ENISA Taxonomy, and VERIS improves standardization and investigation accuracy. Tools such as Wazuh, VirusTotal, AlienVault OTX, and TheHive help analysts perform efficient threat analysis, IOC validation, and incident management in modern cybersecurity environments.





## 4. Basic Incident Response

### Introduction

Basic Incident Response is a structured cybersecurity process used by Security Operations Centers (SOCs) to identify, analyze, contain, eradicate, and recover from cybersecurity incidents. In modern organizations, cyberattacks such as ransomware, phishing, brute-force attacks, malware infections, and unauthorized access attempts occur frequently. Without a proper incident response process, these attacks can cause severe financial loss, service disruption, data breaches, and reputational damage.

Incident response helps organizations:

- minimize attack impact
- reduce downtime
- improve recovery speed
- preserve digital evidence
- improve future security posture
- maintain business continuity

Security teams follow standardized frameworks such as:

- NIST SP 800-61
- SANS Incident Response Framework

These frameworks define the complete incident response lifecycle and help organizations respond to security incidents professionally and systematically.

The incident response lifecycle contains six major phases:

1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

Each phase plays an important role in ensuring that security incidents are handled efficiently and that attackers are removed completely from the environment.

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### Objectives of Basic Incident Response



The primary objectives are:

- identify cyberattacks quickly
- contain malicious activity
- remove threats completely
- recover systems safely
- preserve digital evidence
- improve future detection capabilities
- reduce business impact

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## Incident Response Lifecycle

### 1. Preparation Phase

The preparation phase focuses on building readiness before security incidents occur.

Organizations prepare:

- incident response policies
- SIEM tools
- logging systems
- communication plans
- forensic tools
- backup solutions
- response playbooks

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### Activities Performed During Preparation

- Installing Wazuh SIEM
- Configuring security monitoring
- Creating escalation workflows
- Defining response procedures
- Training SOC analysts
- Maintaining backups
- Configuring firewall rules

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### Example

Installing and configuring Wazuh on Ubuntu Server before attack simulation.



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## 2. Identification Phase

The identification phase focuses on detecting suspicious activity and confirming whether an incident has occurred.

SOC analysts monitor:

- SIEM alerts
- firewall logs
- authentication logs
- endpoint alerts
- suspicious network traffic

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### Indicators of Compromise (IOCs)

Examples:

- suspicious IP addresses
- malware file hashes
- failed login attempts
- abnormal outbound traffic
- suspicious PowerShell activity

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### Example

Wazuh detects multiple failed SSH login attempts from a suspicious attacker IP.

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## 3. Containment Phase

Containment prevents attackers from spreading further across systems or networks.

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### Containment Activities

- isolating infected systems



- disconnecting network access
- blocking malicious IP addresses
- disabling compromised accounts
- stopping malicious processes

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## **Types of Containment**

### Short-Term Containment

Immediate isolation of affected systems.

### Long-Term Containment

Applying permanent fixes and patches.

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### Example

Blocking attacker IP using CrowdSec firewall rules.

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## 4. Eradication Phase

The eradication phase removes malware, attacker persistence mechanisms, and malicious artifacts completely.

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## **Containment Activities**

- removing malware files
- deleting backdoors
- uninstalling malicious software
- patching vulnerabilities
- resetting compromised credentials

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### Example

Removing ransomware executable:



crypto\_locker.exe

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## 5. Recovery Phase

The recovery phase restores systems safely back into production.

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### Activities

- restoring backups
- reconnecting systems
- monitoring for reinfection
- verifying system integrity
- testing applications

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### Example

Restoring an Ubuntu server after malware removal and security patching.

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## 6. Lessons Learned Phase

This phase focuses on improving future incident response processes.

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### Activities

- conducting post-mortem analysis
- identifying root causes
- updating detection rules
- improving SOC playbooks
- improving employee awareness

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### Example

Updating Wazuh detection rules after identifying new brute-force attack patterns.



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## Incident Response Procedures

### System Isolation

System isolation prevents attackers from moving laterally across networks.

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### Isolation Methods

- disconnecting network adapters
- quarantining endpoints
- disabling accounts
- isolating virtual machines

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## Evidence Preservation

Evidence preservation ensures digital evidence remains authentic and untampered.

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### Types of Evidence

|                  |                     |
|------------------|---------------------|
| Evidence Type    | Example             |
| Memory Evidence  | RAM dump            |
| Log Evidence     | Authentication logs |
| File Evidence    | Malware executable  |
| Network Evidence | Packet capture      |

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## Communication Procedures

SOC analysts communicate with:

- management teams
- incident response teams
- system administrators
- legal departments



Professional communication improves coordination during incidents.

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## SOAR Tools

SOAR stands for:

Security Orchestration, Automation, and Response.

SOAR tools automate:

- alert handling
- ticket creation
- IP blocking
- email notifications

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Example SOAR Tool

Splunk Phantom

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## PRACTICAL IMPLEMENTATION – BASIC INCIDENT RESPONSE

Tools Used

Tool| Purpose

Wazuh| Security monitoring

TheHive| Incident management

CrowdSec| IP blocking

Google Docs| Documentation

Draw.io| Workflow diagrams

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### Practical Task 1 – Create Incident Response Template

Objective

Create a professional incident response report.

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## Step 1 – Open Google Docs

Create a new document.

Rename:  
Incident Response Report

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## Step 2 – Add Main Sections

Include:

1. Executive Summary
2. Incident Timeline
3. Impact Analysis
4. Indicators of Compromise
5. Containment Actions
6. Remediation Steps
7. Lessons Learned

---

## Step 3 – Save Document

Export as:

- PDF
- DOCX

---

## Practical Task 2 – Create Investigation Timeline

Objective

Document all response activities in chronological order.

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## Step 1 – Create Timeline Table

Timestamp| Action  
14:00| Alert detected





14:10| Analyzed logs  
14:20| Isolated system  
14:30| Blocked malicious IP  
14:45| Collected evidence

---

Step 2 – Save Timeline

Export as PDF.

---

## Practical Task 3 – Phishing Investigation Checklist

Objective

Investigate phishing attacks systematically.

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Step 1 – Create Checklist

Include:

- Verify email sender
- Analyze email headers
- Validate suspicious links
- Check attachments
- Search VirusTotal
- Identify affected users
- Block malicious domain

---

Step 2 – Save Checklist

Export as PDF.

---

## Practical Task 4 – Wazuh Alert Investigation

Objective



Analyze suspicious alerts in Wazuh.

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Step 1 – Open Wazuh Dashboard

Access:

<https://YOUR-IP>

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Step 2 – Login

Enter administrator credentials.

---

Step 3 – Navigate to Security Events

Open:

- Threat Hunting
- Security Events

---

Step 4 – Analyze Alerts

Check:

- source IP
- failed logins
- suspicious activity
- alert severity
- MITRE ATT&CK mapping

---

Step 5 – Assign Severity

Classify:

- Critical
- High
- Medium



- Low

---

## Practical Task 5 – Containment Activity

### Objective

Contain malicious activity.

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### Step 1 – Identify Attacker IP

Example:

192.168.1.50

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### Step 2 – Block IP Using CrowdSec

Run:

sudo cscli decisions add --ip 192.168.1.50

---

### Step 3 – Verify Blocking

Run:

ping TARGET-IP

---

## Practical Task 6 – Evidence Preservation

### Objective

Preserve digital evidence properly.

---

### Step 1 – Collect Logs

Save:



- authentication logs
- Wazuh alerts
- system logs

---

## Step 2 – Generate SHA256 Hash

Run:

```
sha256sum suspiciousfile.exe
```

---

## Step 3 – Document Evidence

| Evidence           | Hash   | Collected By |
|--------------------|--------|--------------|
| suspiciousfile.exe | SHA256 | SOC Analyst  |

---

## Practical Task 7 – Post-Mortem Report

### Objective

Document lessons learned after incident resolution.

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## Step 1 – Create Summary

Include:

- root cause
- business impact
- response effectiveness
- recommendations

---

## Step 2 – Add Improvements

Examples:

- stronger password policy
- improved firewall rules



- better SIEM monitoring

---

## Expected Outcomes

After completing this practical, the SOC analyst should be able to:

- understand incident response lifecycle
- investigate incidents
- contain attacks
- preserve evidence
- create professional reports
- improve SOC operations

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## Conclusion

Basic Incident Response is an essential cybersecurity process that helps organizations identify, contain, eradicate, and recover from cyberattacks effectively. Using tools such as Wazuh, CrowdSec, TheHive, and proper forensic techniques improves investigation efficiency and strengthens organizational security posture.

## 5. Alert Management Practice

### Introduction

Alert Management is one of the most important responsibilities of a Security Operations Center (SOC). Modern organizations generate thousands of alerts every day from SIEM tools, intrusion detection systems, firewalls, antivirus software, endpoint protection platforms, and cloud security systems. SOC analysts must continuously monitor, classify, prioritize, investigate, and respond to these alerts to protect organizational infrastructure from cyber threats.

Without proper alert management, organizations may experience:

- alert fatigue
- missed critical threats
- delayed response
- security breaches
- operational disruption

Alert management helps security teams:



- identify dangerous threats quickly
- prioritize incidents efficiently
- improve response time
- reduce false positives
- automate investigations
- improve SOC workflows

In this practical implementation, tools such as:

- Wazuh
- TheHive
- Google Sheets
- VirusTotal
- AlienVault OTX

are used to simulate real-world SOC alert handling processes.

The practical tasks focus on:

- alert classification
- severity assignment
- MITRE ATT&CK mapping
- CVSS scoring
- dashboard monitoring
- incident ticket creation
- escalation procedures

This practical provides hands-on experience in real SOC monitoring and alert investigation workflows.

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## Objectives of Alert Management

The primary objectives are:

- classify alerts correctly
- prioritize incidents efficiently
- reduce alert fatigue
- improve investigation workflows
- improve threat visibility
- automate response processes
- improve incident handling

---



## Core Concepts of Alert Management

What is an Alert?

An alert is a notification generated when suspicious or malicious activity is detected.

Alerts are generated from:

- SIEM systems
- firewalls
- IDS/IPS
- endpoint detection tools
- antivirus software
- authentication systems

---

Examples of Security Alerts

| Alert Type           | Description                    |
|----------------------|--------------------------------|
| SSH Brute Force      | Multiple failed login attempts |
| Malware Detection    | Malicious file execution       |
| Port Scan            | Network reconnaissance         |
| Phishing Email       | Suspicious email detected      |
| Privilege Escalation | Unauthorized admin access      |

---

Alert Lifecycle

The alert lifecycle includes:

1. Alert Generation
2. Alert Collection
3. Alert Analysis
4. Alert Prioritization
5. Incident Response
6. Documentation
7. Closure

---

## Alert Severity Levels



SOC analysts classify alerts into:

Severity| Meaning

Critical| Immediate danger

High| Serious threat

Medium| Suspicious activity

Low| Informational activity

---

## CVSS Scoring in Alert Management

CVSS stands for:

Common Vulnerability Scoring System.

It is used to calculate vulnerability severity.

---

## CVSS Severity Table

Score| Severity

0.0 – 3.9| Low

4.0 – 6.9| Medium

7.0 – 8.9| High

9.0 – 10.0| Critical

---

## Example

Vulnerability| CVSS| Severity

Log4Shell| 9.8| Critical

SSH Weak Password| 5.3| Medium

---

## MITRE ATT&CK Mapping

MITRE ATT&CK helps SOC analysts identify attacker tactics and techniques.

---

## Common MITRE Techniques





Technique ID| Description

T1566| Phishing

T1110| Brute Force

T1046| Network Scanning

T1190| Public-Facing Exploit

---

Tools Used in Alert Management

Tool| Purpose

Wazuh| SIEM monitoring

TheHive| Incident management

Google Sheets| Alert classification

VirusTotal| IOC analysis

AlienVault OTX| Threat intelligence

---

## PRACTICAL IMPLEMENTATION – ALERT MANAGEMENT

### Practical Task 1 – Create Alert Classification System

Objective

Create a system to classify alerts based on severity and MITRE ATT&CK techniques.

---

Step 1 – Open Google Sheets

Create a new spreadsheet.

Rename:

SOC Alert Classification

---

Step 2 – Create Table Columns

Add:

Alert ID| Alert Type| Priority| CVSS Score| MITRE Technique| Status

---



## Step 3 – Add Sample Alerts

Example:

| Alert ID | Alert Type        | Priority | CVSS Score | MITRE Technique | Status        |
|----------|-------------------|----------|------------|-----------------|---------------|
| 001      | Log4Shell Exploit | Critical | 9.8        | T1190           | Open          |
| 002      | SSH Brute Force   | Medium   | 5.5        | T1110           | Open          |
| 003      | Phishing Email    | High     | 8.0        | T1566           | Investigating |
| 004      | Port Scan         | Low      | 2.1        | T1046           | Closed        |

---

## Step 4 – Save Documentation

Take screenshots of:

- spreadsheet
- alert table
- priority classification

Export as PDF.

---

## Practical Task 2 – Alert Prioritization

Objective

Prioritize alerts using severity and business impact.

---

### Step 1 – Analyze Alert Severity

Check:

- CVSS score
- exploit availability
- business impact
- affected systems

---

### Step 2 – Identify Asset Criticality



Examples:

- production server = critical
- test VM = low

---

Step 3 – Assign Alert Priority

Assign:

- Critical
- High
- Medium
- Low

based on investigation findings.

---

## Practical Task 3 – Wazuh Dashboard Monitoring

Objective

Monitor alerts using Wazuh SIEM.

---

Step 1 – Open Wazuh Dashboard

Access:

<https://YOUR-IP>

---

Step 2 – Login

Enter:

- username
- password

---



## Step 3 – Open Security Events

Navigate to:

- Threat Hunting
- Security Events
- Dashboard

---

## Step 4 – Analyze Alerts

Check:

- failed logins
- brute-force attempts
- suspicious authentication
- malware detections

---

## Step 5 – Create Dashboard Visualization

Create:

- pie chart
- bar graph

to visualize:

- Critical alerts
- High alerts
- Medium alerts
- Low alerts

---

## Practical Task 4 – Incident Ticket Creation in TheHive

### Objective

Create and manage security incident tickets.

---



## Step 1 – Open TheHive

Access:

http://YOUR-IP:9000

---

## Step 2 – Login

Enter administrator credentials.

---

## Step 3 – Create New Case

Click:

Create Case

---

## Step 4 – Add Incident Details

Title:

[Critical] Ransomware Detected on Server-X

Description:

Indicators detected:

- crypto\_locker.exe
- suspicious encryption activity
- malicious outbound traffic

Priority:

Critical

Assignee:

SOC Analyst

---

## Step 5 – Save Incident Ticket

Click:

Create



---

## Practical Task 5 – Threat Intelligence Validation

### Objective

Validate Indicators of Compromise (IOCs).

---

### Step 1 – Open VirusTotal

Search:

- suspicious IP
- malicious file hash
- phishing domain

---

### Step 2 – Analyze Results

Check:

- antivirus detections
- community score
- malicious activity

---

### Step 3 – Open AlienVault OTX

Search attacker IP.

---

### Step 4 – Review Threat Intelligence

Check:

- attack history
- malware association
- IOC reputation

---



## Practical Task 6 – Escalation Procedure

### Objective

Escalate critical incidents to Tier 2 analysts.

---

### Step 1 – Create Escalation Email

#### Subject:

Critical Alert Escalation – Possible Ransomware Activity

---

### Step 2 – Add Incident Summary

#### Include:

- alert severity
- attacker IP
- affected systems
- IOCs detected
- containment status

---

### Example Escalation Email

A critical ransomware-related alert has been detected on Server-X. Multiple suspicious encryption activities and malicious outbound connections were identified. Initial analysis detected the executable crypto\_locker.exe communicating with IP address 192.168.1.50. Immediate investigation and containment actions are recommended to prevent lateral movement and data loss.

---

## Practical Task 7 – Documentation and Reporting

### Objective

Create professional SOC documentation.

---



## Step 1 – Create Incident Report

Include:

- executive summary
- timeline
- IOC details
- severity
- recommendations

---

## Step 2 – Export Documentation

Save as:

- PDF
- DOCX

Upload to GitHub repository.

---

## Expected Outcomes

After completing this practical, the SOC analyst should be able to:

- classify alerts effectively
- prioritize threats correctly
- use CVSS scoring
- monitor alerts using Wazuh
- create incident tickets in TheHive
- validate IOCs
- perform escalation procedures
- create professional SOC reports

---

## Conclusion

Alert Management is a critical SOC process that helps organizations identify, prioritize, investigate, and respond to cybersecurity threats efficiently. Using tools such as Wazuh, TheHive, VirusTotal, and AlienVault OTX improves SOC visibility, reduces alert fatigue, and strengthens incident response operations in enterprise environments.





## 6. Log Analysis

### Introduction

Log Analysis is the process of collecting, monitoring, and analyzing logs generated from operating systems, applications, firewalls, servers, and security tools. In a Security Operations Center (SOC), log analysis helps analysts identify suspicious activities, detect cyberattacks, investigate incidents, and monitor system behavior in real time.

Logs provide important information such as:

- who performed an action
- when the action occurred
- source and destination IP addresses
- failed or successful login attempts
- malware activities
- network connections

Without proper log analysis, organizations may fail to detect cyber threats, insider attacks, unauthorized access, or malware infections.

---

### Types of Logs

#### 1. System Logs

These logs contain operating system events and activities.

#### 2. Authentication Logs

These logs record login and logout attempts.

#### 3. Firewall Logs

These logs contain allowed and blocked network traffic details.

#### 4. Application Logs

These logs store events generated by software applications.

#### 5. Web Server Logs

These logs record website access requests and responses.



---

## Practical Knowledge

Viewing Authentication Logs in Linux

View Authentication Logs

```
sudo cat /var/log/auth.log
```

Monitor Logs in Real-Time

```
sudo tail -f /var/log/auth.log
```

---

## Detect Failed Login Attempts

Generate Failed SSH Login

```
ssh wronguser@<target-ip>
```

Enter the wrong password multiple times.

Search Failed Login Attempts

```
sudo grep "Failed password" /var/log/auth.log
```

---

## Analyze Logs Using Wazuh

Open the Wazuh Dashboard and go to:

- Security Events
- Discover
- Alerts

Search High Severity Alerts

```
rule.level: > 10
```

Search Authentication Failures

```
authentication failure
```



---

Analyze Apache Logs

View Apache Access Logs

```
sudo tail -f /var/log/apache2/access.log
```

Example Suspicious Request

```
GET /index.php?id=' OR '1'='1
```

---

Skills Learned

- Log monitoring
- Attack detection
- Security investigation
- SIEM analysis
- Real-time monitoring

---

Tools Used

- Wazuh
- Kali Linux
- Apache Server
- Wireshark

## 7. Threat Intelligence

### Introduction

Threat Intelligence is the process of collecting, analyzing, and understanding information related to cyber threats, attackers, malware, vulnerabilities, and attack techniques. In a Security Operations Center (SOC), threat intelligence helps security analysts predict, detect, and prevent cyberattacks before they cause major damage.



Threat intelligence provides information about:

- malicious IP addresses
- phishing domains
- malware signatures
- ransomware campaigns
- attacker behavior
- vulnerabilities and exploits

Organizations use threat intelligence to strengthen their security systems and improve incident response capabilities.

Without threat intelligence, organizations may fail to recognize new cyber threats, advanced attacks, or attacker techniques being used globally.

---

## Types of Threat Intelligence

### 1. Strategic Threat Intelligence

Provides high-level information for management and decision-makers.

Focus:

- cyberattack trends
- business risks
- industry threats

---

### 2. Tactical Threat Intelligence

Provides information about attacker techniques and methods.

Focus:

- phishing attacks
- malware behavior
- attack vectors

---

### 3. Operational Threat Intelligence



Provides information about ongoing attacks and campaigns.

Focus:

- ransomware groups
- hacker activities
- targeted attacks

---

#### **4. Technical Threat Intelligence**

Provides technical indicators used in attacks.

Examples:

- malicious IP addresses
- suspicious domains
- file hashes
- URLs

---

#### **Indicators of Compromise (IOCs)**

Indicators of Compromise are suspicious artifacts that indicate possible cyberattacks.

Examples:

- malicious IP addresses
- phishing links
- malware hashes
- suspicious domains
- abnormal network traffic

---

#### **Threat Intelligence Lifecycle**

1. Data Collection
2. Processing
3. Analysis
4. Threat Identification
5. Reporting
6. Response & Improvement



---

## Practical Knowledge

### Practical 1: Check Malicious IP Address

Use VirusTotal

Open:

- VirusTotal website

Search:

- suspicious IP
- file hash
- URL
- domain

---

### Practical 2: Analyze File Hash in Linux

Generate SHA256 Hash

```
sha256sum suspiciousfile.exe
```

---

Compare Hash Using VirusTotal

Copy the generated hash and search it on VirusTotal.

---

### Practical 3: Monitor Threat Intelligence in Wazuh

Open:

- Wazuh Dashboard
- Security Events

Check for:



- malicious IP alerts
- malware detection
- threat intelligence matches
- blacklist alerts

---

Search High Severity Threats

rule.level: > 12

---

Search Malware Events

malware

---

## Practical 4: Detect Suspicious Network Connections

Use Netstat Command

netstat -antp

This command displays:

- active network connections
- listening ports
- suspicious remote IPs

---

## Practical 5: Capture Traffic Using Wireshark

Open Wireshark and monitor:

- unusual outbound traffic
- unknown protocols
- repeated DNS requests
- suspicious IP communication

---

Common Threat Intelligence Sources



- VirusTotal
- AbuseIPDB
- AlienVault OTX
- MITRE ATT&CK
- OpenPhish

---

## Skills Learned

- Threat detection
- IOC analysis
- Malware investigation
- Network monitoring
- Threat hunting basics

---

## Tools Used

- Wazuh
- Kali Linux
- Wireshark
- VirusTotal
- Netstat

## 8. Malware Analysis Basics

### Introduction

Malware Analysis is the process of studying malicious software to understand its behavior, purpose, attack method, and impact on a system. Malware analysts examine suspicious files and programs to identify how the malware works and how to remove or prevent it.

In a Security Operations Center (SOC), malware analysis helps analysts:

- identify malicious files
- understand attacker techniques
- detect ransomware activity
- investigate infections
- improve incident response





Malware can:

- steal sensitive data
- encrypt files
- spy on users
- damage systems
- provide unauthorized access to attackers

Without malware analysis, organizations may fail to understand how infections spread or how attackers compromise systems.

---

## Types of Malware

### 1. Virus

A malicious program that attaches itself to files and spreads when executed.

---

### 2. Worm

A self-replicating malware that spreads automatically through networks.

---

### 3. Trojan

Malware disguised as legitimate software.

---

### 4. Ransomware

Malware that encrypts files and demands payment for decryption.

---

### 5. Spyware

Malware used to secretly monitor user activities.

---



## 6. Adware

Malware that displays unwanted advertisements.

---

## 7. Rootkit

Malware designed to hide malicious activities from the operating system.

---

## Malware Analysis Types

### 1. Static Analysis

Analyzing malware without executing it.

Focus:

- file hashes
- strings
- file type
- suspicious code

---

### 2. Dynamic Analysis

Analyzing malware while it is running in a controlled environment.

Focus:

- process activity
- network connections
- file changes
- registry changes

---

Indicators of Malware Infection

- slow system performance
- unknown processes
- unusual network traffic



- disabled antivirus
- pop-up advertisements
- encrypted files
- unauthorized access

---

## Practical Knowledge

### Practical 1: Generate File Hash

Generate MD5 Hash

```
md5sum suspiciousfile.exe
```

---

Generate SHA256 Hash

```
sha256sum suspiciousfile.exe
```

Hashes help identify whether a file is malicious.

---

### Practical 2: Analyze File Using VirusTotal

Open:

- VirusTotal website

Upload:

- suspicious file
- or
- search file hash

Analyze:

- antivirus detections
- malware family
- community reports
- behavior analysis

---



## Practical 3: Extract Strings from Suspicious File

Use Strings Command

```
strings suspiciousfile.exe
```

This command extracts readable text from the malware file.

Look for:

- IP addresses
- URLs
- commands
- suspicious messages

---

## Practical 4: Monitor Running Processes

View Active Processes

```
ps aux
```

Check for:

- unknown processes
- suspicious applications
- abnormal CPU usage

---

## Practical 5: Analyze Network Connections

View Active Connections

```
netstat -antp
```

This command shows:

- active network sessions
- listening ports
- suspicious external IPs

---



## Practical 6: Monitor Malware Activity Using Wireshark

Open Wireshark and monitor:

- suspicious outbound traffic
- DNS requests
- malware communication
- unusual protocols

Analyze packets to identify:

- command and control traffic
- data exfiltration
- malicious connections

---

## Practical 7: Malware Detection in Wazuh

Open:

- Wazuh Dashboard
- Security Events
- Alerts

Search for:

- malware alerts
- suspicious processes
- malicious file detections
- abnormal behavior

---

Search Malware Events

malware

---

Search High Severity Alerts

rule.level: > 12



---

## Basic Malware Analysis Workflow

1. Collect Suspicious File
2. Generate File Hash
3. Scan File in VirusTotal
4. Perform Static Analysis
5. Monitor Malware Behavior
6. Analyze Network Activity
7. Document Findings

---

## Skills Learned

- malware detection
- file hash analysis
- process monitoring
- network traffic analysis
- basic forensic investigation

---

## Tools Used

- Wazuh
- Kali Linux
- Wireshark
- VirusTotal
- Netstat
- Strings

## 9. Vulnerability Assessment

### Introduction

Vulnerability Assessment is the process of identifying, analyzing, and evaluating security weaknesses present in systems, networks, applications, and devices. In cybersecurity, vulnerabilities are weaknesses that attackers can exploit to gain unauthorized access, steal data, or damage systems.



In a Security Operations Center (SOC), vulnerability assessment helps security teams:

- identify security weaknesses
- reduce attack surfaces
- improve system security
- prevent cyberattacks
- prioritize security fixes

Organizations perform vulnerability assessments regularly to detect outdated software, weak passwords, open ports, misconfigurations, and insecure services before attackers exploit them.

Without vulnerability assessment, organizations may remain exposed to cyber threats such as:

- ransomware attacks
- unauthorized access
- data breaches
- malware infections
- network compromise

---

## Types of Vulnerabilities

### 1. Software Vulnerabilities

Weaknesses found in operating systems or applications.

Examples:

- outdated software
- missing security patches
- insecure applications

---

### 2. Network Vulnerabilities

Weaknesses present in network devices or configurations.

Examples:

- open ports
- weak firewall rules



- insecure protocols

---

### 3. Configuration Vulnerabilities

Improper security settings in systems.

Examples:

- default passwords
- unnecessary services enabled
- weak authentication settings

---

### 4. Web Application Vulnerabilities

Weaknesses found in websites or web applications.

Examples:

- SQL Injection
- Cross-Site Scripting (XSS)
- Broken Authentication

---

### Vulnerability Assessment Process

1. Asset Identification
2. Vulnerability Scanning
3. Risk Analysis
4. Vulnerability Validation
5. Reporting
6. Remediation

---

### Vulnerability Severity Levels

Critical

Requires immediate action because it can severely impact systems.





---

## High

Serious vulnerability that attackers can exploit easily.

---

## Medium

Moderate risk vulnerability.

---

## Low

Minor vulnerability with limited impact.

---

## Practical Knowledge

### Practical 1: Scan Open Ports Using Nmap

#### Basic Network Scan

`nmap <target-ip>`

This command identifies:

- open ports
- active services
- running systems

---

#### Service Version Detection

`nmap -sV <target-ip>`

This command detects:

- service versions
- software information



---

## Practical 2: Vulnerability Scan Using Nmap Scripts

Run Vulnerability Detection Scripts

```
nmap --script vuln <target-ip>
```

This command scans for:

- known vulnerabilities
- weak services
- insecure configurations

---

## Practical 3: Scan Web Application Vulnerabilities

Nikto Web Scan

```
nikto -h http://<target-ip>
```

Nikto checks for:

- outdated web servers
- insecure files
- dangerous configurations
- known web vulnerabilities

---

## Practical 4: Analyze Vulnerabilities in Wazuh

Open:

- Wazuh Dashboard
- Vulnerability Detection
- Security Events

Check for:

- outdated software
- missing patches
- vulnerable applications
- CVE alerts



---

Search Vulnerability Alerts

vulnerability

---

Search High Severity Vulnerabilities

rule.level: > 10

---

## Practical 5: Identify Weak Passwords

Use Hydra for Password Testing

```
hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://<target-ip>
```

This tests:

- weak passwords
- insecure authentication

---

## Practical 6: Analyze Active Services

View Listening Ports

```
netstat -tulnp
```

This command displays:

- listening services
- open network ports
- suspicious services

---

## Common Vulnerability Examples

SQL Injection



Attackers inject malicious SQL queries into applications.

---

Cross-Site Scripting (XSS)

Attackers inject malicious scripts into websites.

---

Weak Passwords

Simple passwords that attackers can guess or crack easily.

---

Open Ports

Unused open ports increase attack surface.

---

Outdated Software

Old software may contain known vulnerabilities.

---

Basic Vulnerability Assessment Workflow

1. Identify Target System
2. Scan Open Ports
3. Detect Services
4. Run Vulnerability Scan
5. Analyze Results
6. Prioritize Risks
7. Apply Security Fixes

---

Skills Learned

- vulnerability scanning
- network analysis



- web application testing
- service enumeration
- risk assessment

---

## Tools Used

- Nmap
- Nikto
- Wazuh
- Hydra
- Kali Linux
- Netstat

## 10. Network Traffic Monitoring

### Introduction

Network Traffic Monitoring is the process of capturing, analyzing, and monitoring data packets flowing through a network. In cybersecurity, network traffic monitoring helps security analysts detect suspicious activities, identify cyberattacks, monitor bandwidth usage, and investigate security incidents.

In a Security Operations Center (SOC), analysts continuously monitor network traffic to identify:

- unauthorized access
- malware communication
- suspicious IP addresses
- abnormal network behavior
- data exfiltration
- brute-force attacks

Without proper network monitoring, organizations may fail to detect attacks occurring inside the network.



---

## Importance of Network Traffic Monitoring

Network monitoring helps organizations:

- detect cyberattacks early
- monitor suspicious traffic
- identify infected systems
- prevent data theft
- improve incident response
- analyze attacker behavior

---

## Types of Network Traffic

### 1. Inbound Traffic

Traffic entering the network from external sources.

Example:

- website requests
- remote connections

---

### 2. Outbound Traffic

Traffic leaving the network.

Example:

- DNS requests
- file uploads
- malware communication

---

### 3. Internal Traffic

Traffic between devices inside the same network.



Example:

- communication between servers
- internal file sharing

---

Common Indicators of Suspicious Traffic

- unusual outbound connections
- repeated DNS requests
- communication with unknown IPs
- large data transfers
- abnormal port activity
- excessive failed connections
- traffic spikes

---

Network Monitoring Protocols

HTTP/HTTPS

Used for web traffic communication.

---

DNS

Used for domain name resolution.

---

TCP

Used for reliable communication between devices.

---

UDP

Used for fast communication with low overhead.

---



## ICMP

Used for network diagnostics and ping requests.

---

## Practical Knowledge

### Practical 1: Capture Packets Using Wireshark

#### Open Wireshark

Select the active network interface and start packet capture.

Monitor:

- incoming traffic
- outgoing traffic
- suspicious connections
- DNS requests

---

### Practical 2: Filter Traffic in Wireshark

#### Filter HTTP Traffic

http

---

#### Filter DNS Traffic

dns

---

#### Filter Traffic by IP Address

ip.addr == <target-ip>

---

#### Filter TCP Traffic





tcp

---

## Practical 3: Analyze Active Connections

### View Active Network Connections

netstat -antp

This command shows:

- active connections
- listening ports
- connected remote IPs

---

## Practical 4: Monitor Network Interfaces

### Display Network Interfaces

ifconfig

or

ip a

---

## Practical 5: Capture Packets Using Tcpdump

### Start Packet Capture

sudo tcpdump -i eth0

---

### Capture Traffic from Specific IP

sudo tcpdump host <target-ip>

---

## Practical 6: Analyze Network Alerts in Wazuh



Open:

- Wazuh Dashboard
- Security Events
- Network Monitoring Alerts

Check for:

- suspicious IP activity
- brute-force attacks
- malware communication
- unusual traffic behavior

---

Search High Severity Alerts

rule.level: > 10

---

Search Network Events

network

---

Common Network Attacks Detected Through Monitoring

## 1. Brute-Force Attack

Multiple failed login attempts from attacker IPs.

---

## 2. DDoS Attack

Large amount of traffic sent to overwhelm systems.

---

## 3. Malware Communication

Infected systems communicating with attacker servers.



---

#### 4. Port Scanning

Attackers scanning systems for open ports.

---

#### 5. Data Exfiltration

Unauthorized transfer of sensitive data outside the network.

---

#### Basic Network Monitoring Workflow

1. Capture Network Traffic
2. Analyze Packets
3. Detect Suspicious Activity
4. Identify Threat Source
5. Investigate Incident
6. Generate Alerts
7. Respond to Threat

---

#### Skills Learned

- packet analysis
- network monitoring
- traffic filtering
- attack detection
- suspicious connection analysis

---

#### Tools Used

- Wireshark
- Tcpdump
- Wazuh
- Netstat
- Kali Linux



## 11. SIEM Dashboard Monitoring

### Introduction

SIEM (Security Information and Event Management) Dashboard Monitoring is the process of continuously monitoring security alerts, logs, events, and incidents through a centralized dashboard. SIEM tools collect logs from different devices, servers, applications, firewalls, and security tools to help analysts detect and respond to cyber threats in real time.

In a Security Operations Center (SOC), SIEM dashboards are used to:

- monitor security alerts
- investigate suspicious activities
- detect cyberattacks
- analyze logs
- prioritize incidents
- improve threat detection

SIEM dashboards provide a centralized view of the organization's security environment.

Without SIEM monitoring, organizations may fail to detect attacks occurring across multiple systems.

---

### Functions of SIEM Dashboard

#### 1. Log Collection

Collects logs from:

- servers
- firewalls
- routers
- applications



- endpoints

---

## 2. Event Correlation

Connects related events to identify attacks.

Example:

- multiple failed logins
- malware alerts
- suspicious IP communication

---

## 3. Real-Time Monitoring

Continuously monitors security events and alerts.

---

## 4. Alert Generation

Generates alerts when suspicious activity is detected.

---

## 5. Incident Investigation

Helps analysts investigate attacks using logs and events.

---

### Common SIEM Alerts

- failed login attempts
- brute-force attacks
- malware detection
- privilege escalation
- suspicious network traffic
- unauthorized access
- policy violations

---



## Alert Severity Levels

### Critical

Immediate response required.

---

### High

Serious threat that may impact systems.

---

### Medium

Moderate suspicious activity.

---

### Low

Minor activity with low risk.

---

## Practical Knowledge

### Practical 1: Open Wazuh Dashboard

Login to:

- Wazuh Dashboard

Navigate to:

- Security Events
- Discover
- Dashboard
- Alerts

Monitor:

- active alerts



- event severity
- suspicious activities

---

## Practical 2: Search High Severity Alerts

### Search Critical Alerts

rule.level: > 12

---

### Search Medium and High Alerts

rule.level: >= 7

---

## Practical 3: Search Authentication Failures

### Search Failed Login Attempts

authentication failure

Check:

- source IP
- username
- timestamp
- targeted system

---

## Practical 4: Monitor Malware Alerts

### Search Malware Events

malware

Analyze:

- affected system
- malware name
- alert severity



- event details

---

## Practical 5: Monitor Network Alerts

### Search Network Events

network

Check for:

- suspicious connections
- unusual traffic
- port scanning
- brute-force attacks

---

## Practical 6: Investigate Alert Details

Open an alert and analyze:

- event description
- rule ID
- MITRE ATT&CK mapping
- source IP
- destination IP
- hostname
- timestamp

---

## Practical 7: Create Alert Filters

Filter alerts based on:

- severity
- hostname
- IP address
- attack type
- event category

Example:





agent.name: kali

---

## SIEM Monitoring Workflow

1. Collect Logs
2. Normalize Events
3. Correlate Activities
4. Generate Alerts
5. Investigate Incidents
6. Escalate Serious Threats
7. Respond and Document

---

## Benefits of SIEM Monitoring

- centralized monitoring
- faster threat detection
- improved incident response
- log management
- real-time visibility
- attack investigation

---

## Skills Learned

- SIEM monitoring
- alert investigation
- log correlation
- incident analysis
- dashboard monitoring

---

## Tools Used

- Wazuh
- Kali Linux
- Wireshark
- Elasticsearch
- Kibana



## 12. Use Cases and Correlation Rules

### Introduction

Use Cases and Correlation Rules are important components of a SIEM (Security Information and Event Management) system. In a Security Operations Center (SOC), they help analysts detect suspicious activities, cyberattacks, and abnormal behavior automatically.

A SIEM tool collects logs from multiple systems such as:

- servers
- firewalls
- endpoints
- applications
- network devices

Correlation rules analyze these logs and connect related events to identify possible attacks.

For example:

- multiple failed login attempts
- login success after several failures
- malware detection from multiple systems
- unusual outbound traffic



These events may individually appear harmless, but when correlated together, they can indicate an ongoing attack.

---

What is a Use Case?

A Use Case is a real-world security scenario created to detect suspicious activities or cyberattacks.

Use cases help organizations:

- identify attacks quickly
- automate detection
- improve monitoring
- reduce manual investigation
- strengthen incident response

---

Examples of SIEM Use Cases

## 1. Brute-Force Detection

Detects repeated failed login attempts.

Indicators:

- multiple authentication failures
- repeated login attempts from same IP

---

## 2. Malware Detection

Detects malicious file activity or suspicious processes.

Indicators:

- malware alerts
- suspicious executables
- abnormal system behavior

---



### 3. Privilege Escalation Detection

Detects unauthorized administrator access.

Indicators:

- sudden admin privilege assignment
- unusual account activity

---

### 4. Data Exfiltration Detection

Detects unauthorized transfer of sensitive data.

Indicators:

- large outbound traffic
- unusual external connections

---

### 5. Port Scanning Detection

Detects attackers scanning systems for open ports.

Indicators:

- connection attempts to multiple ports
- repeated network probes

---

### What are Correlation Rules?

Correlation Rules are logical conditions created inside SIEM tools to detect suspicious activities automatically.

These rules:

- analyze logs
- compare events
- identify attack patterns
- generate alerts



Example:

If:

- more than 10 failed logins occur within 5 minutes

Then:

- generate brute-force attack alert

---

## Types of Correlation Rules

### 1. Threshold-Based Rules

Generate alerts after a specific limit is reached.

Example:

- more than 20 failed logins

---

### 2. Behavioral Rules

Detect abnormal user or system behavior.

Example:

- login at unusual hours

---

### 3. Pattern-Based Rules

Detect known attack patterns.

Example:

- malware activity followed by outbound traffic

---

### 4. Event Chaining Rules



Connect multiple related events together.

Example:

- failed login → successful login → privilege escalation

---

Practical Knowledge

Practical 1: Monitor Alerts in Wazuh

Open:

- Wazuh Dashboard
- Security Events
- Alerts

Check:

- brute-force alerts
- malware alerts
- suspicious activity
- authentication failures

---

Practical 2: Search Failed Login Attempts

Search Authentication Failures

authentication failure

Analyze:

- source IP
- failed usernames
- number of attempts
- timestamps

---

Practical 3: Detect Brute-Force Activity

Generate Failed SSH Logins



```
ssh wronguser@<target-ip>
```

Enter incorrect passwords multiple times.

---

Analyze Logs

```
sudo grep "Failed password" /var/log/auth.log
```

---

Practical 4: Search High Severity Alerts

Search Critical Alerts

```
rule.level: > 12
```

---

Practical 5: Detect Port Scanning Activity

Run Nmap Scan

```
nmap <target-ip>
```

This simulates:

- port scanning activity
- service enumeration

Wazuh may generate alerts related to scanning behavior.

---

Practical 6: Monitor Network Connections

View Active Connections

```
netstat -antp
```

Check for:

- suspicious remote IPs



- abnormal outbound connections
- repeated connections

---

## Practical 7: Create Correlation Logic

### Example Correlation Rule:

If:

- more than 10 failed logins occur
- from same IP
- within 5 minutes

Then:

- generate brute-force attack alert

---

### Benefits of Correlation Rules

- automated threat detection
- reduced false positives
- faster incident response
- attack pattern identification
- improved SOC efficiency

---

### Basic Workflow

1. Collect Logs
2. Monitor Events
3. Apply Correlation Rules
4. Detect Suspicious Activity
5. Generate Alerts
6. Investigate Incident
7. Respond to Threat

---

### Skills Learned





- SIEM correlation
- attack detection
- alert monitoring
- event analysis
- SOC investigation

---

## Tools Used

- Wazuh
- Kali Linux
- Nmap
- Netstat
- Wireshark



## 13. Incident Documentation and Reporting

### Introduction

Incident Documentation and Reporting is the process of recording, organizing, and reporting all details related to a cybersecurity incident. In a Security Operations Center (SOC), proper documentation helps analysts track incidents, investigate attacks, improve response procedures, and maintain security records.

Every security incident must be documented properly because it helps:

- understand what happened
- identify attack methods
- track response actions
- support future investigations
- improve security measures
- maintain compliance requirements

Without proper documentation, organizations may lose important evidence, fail to understand attacks, or repeat the same security mistakes.

---

### Importance of Incident Documentation

Incident documentation helps organizations:

- maintain investigation records
- improve incident response
- support forensic analysis
- share information with teams
- meet legal and compliance requirements
- create security reports for management

---

### Information Included in Incident Reports

#### 1. Incident ID

Unique identification number for the incident.



---

## 2. Date and Time

Time when the incident was detected.

---

## 3. Incident Type

Type of attack or suspicious activity.

Examples:

- malware infection
- brute-force attack
- phishing attack
- unauthorized access

---

## 4. Affected Systems

Systems, servers, or devices impacted by the incident.

---

## 5. Source IP Address

Attacker or suspicious IP address.

---

## 6. Severity Level

Risk level of the incident.

Examples:

- Critical
- High
- Medium
- Low

---



## 7. Actions Taken

Steps performed during investigation and response.

Examples:

- blocked IP
- isolated system
- removed malware
- reset passwords

---

## 8. Final Resolution

Final outcome of the investigation.

---

## Types of Security Reports

### 1. Incident Report

Detailed report about a specific security incident.

---

### 2. Daily SOC Report

Summary of daily alerts and incidents.

---

### 3. Vulnerability Report

Information about identified vulnerabilities.

---

### 4. Malware Analysis Report

Details about malware behavior and impact.

---



## 5. Compliance Report

Reports created for legal and security standards.

---

### Practical Knowledge

#### Practical 1: Investigate Alert in Wazuh

Open:

- Wazuh Dashboard
- Security Events
- Alerts

Analyze:

- alert description
- severity
- source IP
- affected system
- event timestamp

---

#### Practical 2: Search High Severity Alerts

Search Critical Alerts

rule.level: > 12

---

Search Authentication Failures

authentication failure

---

#### Practical 3: Collect Evidence from Logs

View Authentication Logs



```
sudo cat /var/log/auth.log
```

---

Search Failed Login Attempts

```
sudo grep "Failed password" /var/log/auth.log
```

Collect:

- source IP
- targeted usernames
- timestamps
- attack patterns

---

Practical 4: Document Incident Details

Record:

- incident ID
- attack type
- affected systems
- severity level
- evidence collected
- actions taken
- analyst observations

---

Practical 5: Create Basic Incident Report

Example Incident Report Structure

Incident Title:

SSH Brute-Force Attack

Date:

14 August 2025

Severity:



High

Affected System:

Linux Server

Source IP:

192.168.1.10

Description:

Multiple failed SSH login attempts detected from external IP address.

Actions Taken:

- blocked attacker IP
- reviewed logs
- escalated to Tier 2 analyst

Final Status:

Incident contained successfully.

---

Practical 6: Export Alerts from Wazuh

Open:

- Wazuh Dashboard
- Alerts Section

Export:

- alert logs
- investigation details
- event reports

Formats:

- CSV
- JSON
- PDF



---

## Incident Response Documentation Workflow

1. Detect Incident
2. Investigate Alert
3. Collect Evidence
4. Analyze Logs
5. Document Findings
6. Report Incident
7. Review and Improve

---

## Best Practices for Documentation

- maintain accurate records
- document every action
- include timestamps
- preserve evidence
- use clear language
- avoid missing details

---

## Skills Learned

- incident reporting
- evidence collection
- log documentation
- alert investigation
- report writing

---

## Tools Used

- Wazuh
- Kali Linux
- Linux Logs
- Wireshark
- SIEM Dashboard





## 14. MITRE ATT&CK Basics

### Introduction

MITRE ATT&CK is a globally recognized cybersecurity framework used to understand attacker behavior, attack techniques, and real-world cyberattack methods. ATT&CK stands for:

Adversarial Tactics, Techniques, and Common Knowledge.

In a Security Operations Center (SOC), the MITRE ATT&CK framework helps analysts:

- understand attacker behavior
- identify attack stages
- improve threat detection
- strengthen incident response
- map security alerts to attack techniques

The framework is based on real-world cyberattack observations and is widely used by:

- SOC analysts
- threat hunters
- incident response teams
- cybersecurity researchers

---

### Components of MITRE ATT&CK

#### 1. Tactics

Tactics represent the attacker's objective during an attack.

Examples:

- Initial Access
- Execution
- Persistence



- Privilege Escalation
- Defense Evasion
- Credential Access
- Discovery
- Lateral Movement
- Exfiltration

---

## 2. Techniques

Techniques describe how attackers achieve their objectives.

Example:

- phishing emails
- brute-force attacks
- command execution
- credential dumping

---

## 3. Sub-Techniques

More detailed methods used within techniques.

Example:

- spear phishing attachment
- spear phishing link

---

## Common MITRE ATT&CK Tactics

### Initial Access

How attackers enter systems.

Examples:

- phishing
- exploiting vulnerabilities
- malicious USB devices



---

## Execution

Running malicious code on the victim system.

Examples:

- PowerShell commands
- malware execution
- scripts

---

## Persistence

Maintaining long-term access.

Examples:

- startup services
- scheduled tasks
- backdoors

---

## Privilege Escalation

Gaining higher system privileges.

Examples:

- exploiting system vulnerabilities
- abusing administrator permissions

---

## Defense Evasion

Avoiding detection by security tools.

Examples:

- disabling antivirus
- obfuscating malware



---

## Credential Access

Stealing usernames and passwords.

Examples:

- keylogging
- password dumping

---

## Discovery

Gathering information about systems and networks.

Examples:

- network scanning
- system enumeration

---

## Lateral Movement

Moving across



## 15. Compliance and Security Policies

### Introduction

Compliance and Security Policies are important parts of cybersecurity that help organizations protect systems, networks, applications, and sensitive information. Security policies define the rules, guidelines, and procedures employees and security teams must follow to maintain security.

Compliance ensures that organizations follow legal, regulatory, and industry security standards.

In a Security Operations Center (SOC), compliance and security policies help:

- improve security management
- reduce cyber risks
- protect sensitive data
- maintain legal requirements
- standardize security procedures
- improve incident response

Without proper security policies and compliance standards, organizations may face:

- data breaches
- financial penalties
- legal issues
- security misconfigurations
- weak security controls

---

### What is a Security Policy?

A Security Policy is a document containing rules and procedures designed to protect organizational assets and data.

Security policies define:

- acceptable system usage
- password requirements
- access control rules
- incident response procedures
- employee responsibilities

---



## Types of Security Policies

### 1. Password Policy

Defines password security requirements.

Examples:

- minimum password length
- password complexity
- password expiration rules

---

### 2. Access Control Policy

Defines who can access systems and data.

Examples:

- role-based access
- least privilege principle
- administrator permissions

---

### 3. Incident Response Policy

Defines steps for handling security incidents.

Examples:

- detection procedures
- escalation process
- reporting methods

---

### 4. Acceptable Use Policy

Defines how employees can use company systems and networks.

Examples:



- internet usage
- software installation restrictions
- email usage rules

---

## 5. Data Protection Policy

Defines methods for protecting sensitive data.

Examples:

- encryption
- backup procedures
- secure data storage

---

What is Compliance?

Compliance means following cybersecurity laws, standards, and regulations required by governments or industries.

Organizations must comply with security standards to:

- protect customer data
- improve security
- avoid penalties
- maintain trust

---

## Common Cybersecurity Compliance Standards

### 1. ISO 27001

International standard for Information Security Management Systems (ISMS).

Focus:

- risk management
- security controls
- information protection

---



## 2. GDPR

General Data Protection Regulation for protecting personal data.

Focus:

- privacy protection
- user consent
- data security

---

## 3. HIPAA

Security standard for healthcare information protection.

Focus:

- patient data security
- healthcare privacy

---

## 4. PCI-DSS

Security standard for payment card information.

Focus:

- secure card transactions
- payment data protection

---

## 5. NIST Cybersecurity Framework

Framework for managing cybersecurity risks.

Focus:

- identification
- protection
- detection
- response





- recovery

---

## Practical Knowledge

### Practical 1: Check User Accounts in Linux

#### View User Accounts

```
cat /etc/passwd
```

This helps identify:

- active users
- unauthorized accounts
- system accounts

---

### Practical 2: Check Password Policy

#### View Password Policy Configuration

```
sudo cat /etc/login.defs
```

Analyze:

- password aging
- password expiration
- security configurations

---

### Practical 3: Monitor Failed Login Attempts

#### Search Failed Logins

```
sudo grep "Failed password" /var/log/auth.log
```

This helps monitor:

- unauthorized access attempts
- brute-force attacks
- policy violations



---

## Practical 4: Review Open Ports

### View Listening Ports

```
netstat -tulnp
```

Check for:

- unnecessary services
- unauthorized ports
- insecure applications

---

## Practical 5: Monitor Compliance Alerts in Wazuh

Open:

- Wazuh Dashboard
- Security Events
- Compliance Monitoring

Check:

- policy violations
- suspicious activities
- compliance alerts
- system misconfigurations

---

### Search Compliance Events

```
compliance
```

---

### Search High Severity Alerts

```
rule.level: > 10
```

---



## Security Best Practices

- use strong passwords
- enable multi-factor authentication
- regularly update systems
- monitor logs continuously
- restrict unnecessary access
- encrypt sensitive data
- perform regular backups

---

## Compliance Monitoring Workflow

1. Define Security Policies
2. Implement Security Controls
3. Monitor Systems
4. Detect Violations
5. Investigate Issues
6. Apply Remediation
7. Generate Compliance Reports

---

## Skills Learned

- policy management
- compliance monitoring
- access control analysis
- security auditing
- log monitoring

---

## Tools Used

- Wazuh
- Kali Linux
- Netstat
- Linux Logs
- SIEM Dashboard



## 16. Final SOC Workflow and End-to-End Attack Simulation

### Introduction

A Security Operations Center (SOC) follows a structured workflow to detect, analyze, investigate, respond to, and recover from cybersecurity incidents. The SOC workflow combines monitoring tools, security analysts, incident response procedures, and threat intelligence to protect organizational systems from cyberattacks.

End-to-End Attack Simulation helps analysts understand how real cyberattacks occur and how security teams respond during each attack stage.

In this process, analysts learn:

- attack detection
- log analysis
- alert investigation
- incident escalation
- threat containment
- recovery procedures

This practical workflow combines all previously learned SOC concepts into a complete incident handling process.

---

### SOC Workflow Stages

#### 1. Monitoring

SOC analysts continuously monitor:

- security alerts
- network traffic
- system logs
- suspicious activities

Tools Used:

- SIEM dashboards
- IDS/IPS
- Wazuh
- Wireshark



---

## 2. Alert Detection

Security tools generate alerts when suspicious activities are detected.

Examples:

- failed login attempts
- malware detection
- abnormal traffic
- privilege escalation

---

## 3. Alert Triage

Analysts classify alerts based on:

- severity
- impact
- attack type
- affected systems

Severity Levels:

- Critical
- High
- Medium
- Low

---

## 4. Investigation

SOC analysts investigate:

- source IP address
- logs
- attack behavior
- malware activity
- affected systems

---



## 5. Incident Escalation

Critical incidents are escalated to:

- Tier 2 analysts
- Incident Response Team
- Management Team

---

## 6. Containment

Security teams isolate affected systems to stop the attack from spreading.

Examples:

- blocking malicious IPs
- disabling compromised accounts
- disconnecting infected systems

---

## 7. Eradication

Remove malicious components from systems.

Examples:

- malware removal
- patching vulnerabilities
- deleting malicious files

---

## 8. Recovery

Restore systems back to normal operation.

Examples:

- restoring backups
- restarting services
- validating system integrity

---



## 9. Documentation and Reporting

Document:

- attack details
- affected systems
- actions taken
- investigation findings
- final resolution

---

### End-to-End Attack Simulation

Scenario:

An attacker performs a brute-force SSH attack on a Linux server.

SOC analysts must:

- detect the attack
- investigate logs
- identify attacker IP
- escalate the incident
- block the attacker
- document the incident

---

### Practical Knowledge

#### Practical 1: Simulate Brute-Force Attack

##### Run Hydra Attack

```
sudo hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://<target-ip>
```

This generates:

- multiple failed login attempts
- brute-force attack activity

---



## Practical 2: Monitor Alerts in Wazuh

Open:

- Wazuh Dashboard
- Security Events
- Alerts

Monitor:

- authentication failures
- brute-force alerts
- suspicious IP activity

---

Search High Severity Alerts

rule.level: > 12

---

Search Authentication Failures

authentication failure

---

## Practical 3: Analyze Authentication Logs

View Authentication Logs

```
sudo cat /var/log/auth.log
```

---

Search Failed Login Attempts

```
sudo grep "Failed password" /var/log/auth.log
```

Collect:

- attacker IP
- targeted usernames
- timestamps





- attack frequency

---

## Practical 4: Analyze Network Traffic

### Monitor Traffic Using Wireshark

Check for:

- repeated SSH connections
- suspicious IP communication
- abnormal traffic spikes

---

### Capture Packets Using Tcpdump

```
sudo tcpdump -i eth0
```

---

## Practical 5: Block Attacker IP

### Block IP Using UFW Firewall

```
sudo ufw deny from <attacker-ip>
```

This blocks further attack attempts.

---

## Practical 6: Verify Active Connections

### View Active Network Sessions

```
netstat -antp
```

Check for:

- suspicious active sessions
- unauthorized remote connections

---



## Practical 7: Escalate the Incident

Escalate to:

- Tier 2 Analyst
- Incident Response Team

Provide:

- incident details
- attacker IP
- affected systems
- evidence collected

---

## Practical 8: Document the Incident

Include:

- incident ID
- attack type
- timestamps
- severity level
- actions taken
- final resolution

---

## Complete SOC Investigation Workflow

1. Monitor Alerts
2. Detect Suspicious Activity
3. Investigate Logs
4. Analyze Network Traffic
5. Identify Threat Source
6. Escalate Incident
7. Contain the Attack
8. Remove Threat
9. Recover Systems
10. Document Findings

---

Skills Learned



- SOC monitoring
- attack investigation
- incident response
- malware detection
- network analysis
- security documentation

---

## Tools Used

- Wazuh
- Kali Linux
- Hydra
- Wireshark
- Tcpdump
- Netstat
- UFW Firewall

## 17. Digital Forensics Basics

### Introduction



Digital Forensics is the process of collecting, preserving, analyzing, and investigating digital evidence after a cybersecurity incident or cybercrime. In cybersecurity and SOC environments, digital forensics helps analysts understand how an attack occurred, what systems were affected, and what actions the attacker performed.

Digital forensics is important for:

- incident investigation
- evidence collection
- attack analysis
- malware investigation
- identifying attackers
- legal investigations

The main goal of digital forensics is to preserve evidence without modifying the original data.

Without proper forensic investigation, organizations may lose critical evidence needed for incident response or legal action.

---

## Types of Digital Forensics

### 1. Disk Forensics

Investigates data stored on hard drives and storage devices.

Examples:

- deleted files
- malware files
- hidden data

---

### 2. Network Forensics

Analyzes network traffic and communication.

Examples:

- suspicious connections
- malware traffic
- attacker IP addresses



---

### 3. Memory Forensics

Analyzes RAM memory contents.

Examples:

- running malware
- active processes
- encryption keys

---

### 4. Email Forensics

Investigates suspicious emails and phishing attacks.

Examples:

- phishing emails
- malicious attachments
- spoofed email addresses

---

### 5. Malware Forensics

Investigates malicious software behavior.

Examples:

- ransomware
- trojans
- spyware

---

### Digital Forensics Process

1. Identification
2. Collection
3. Preservation
4. Examination
5. Analysis



6. Documentation

7. Reporting

---

## Chain of Custody

Chain of Custody is a process used to maintain evidence integrity during an investigation.

It records:

- who collected evidence
- when evidence was collected
- where evidence was stored
- who accessed the evidence

This ensures evidence is not modified or tampered with.

---

## Practical Knowledge

### Practical 1: View Running Processes

#### Display Active Processes

`ps aux`

Check for:

- suspicious processes
- unknown applications
- abnormal resource usage

---

### Practical 2: Analyze Active Network Connections

#### View Active Connections

`netstat -antp`

This command shows:

- active sessions



- listening ports
- suspicious remote IP addresses

---

## Practical 3: Collect Authentication Logs

### View Authentication Logs

```
sudo cat /var/log/auth.log
```

Analyze:

- failed logins
- unauthorized access
- suspicious authentication attempts

---

## Practical 4: Search Failed Login Attempts

```
sudo grep "Failed password" /var/log/auth.log
```

Collect:

- attacker IP address
- targeted usernames
- timestamps

---

## Practical 5: Capture Network Traffic

### Capture Packets Using Tcpcap

```
sudo tcpdump -i eth0
```

This captures:

- incoming traffic
- outgoing traffic
- suspicious packets

---



## Practical 6: Analyze Traffic Using Wireshark

Open Wireshark and monitor:

- suspicious communication
- malware traffic
- repeated DNS requests
- abnormal protocols

---

## Practical 7: Generate File Hash

Generate SHA256 Hash

```
sha256sum suspiciousfile.exe
```

Hashes help verify file integrity during investigations.

---

## Practical 8: Investigate Malware File

Extract Strings from File

```
strings suspiciousfile.exe
```

Look for:

- suspicious URLs
- IP addresses
- commands
- malware indicators

---

## Practical 9: Investigate Alerts in Wazuh

Open:

- Wazuh Dashboard
- Security Events
- Alerts

Analyze:





- malware alerts
- authentication failures
- suspicious network events

---

## Search High Severity Alerts

rule.level: > 12

---

## Evidence Collection Best Practices

- never modify original evidence
- document all actions
- maintain timestamps
- preserve file integrity
- securely store evidence
- maintain chain of custody

---

## Basic Forensic Investigation Workflow

1. Detect Incident
2. Collect Evidence
3. Preserve Data
4. Analyze Logs and Traffic
5. Identify Attack Method
6. Document Findings
7. Generate Final Report

---

## Skills Learned

- evidence collection
- log investigation
- network forensics
- malware investigation
- forensic documentation

---



## Tools Used

- Wazuh
- Kali Linux
- Wireshark
- Tcpdump
- Netstat
- Strings